THE DISCOVERY GAP IN AI-MEDIATED MARKETS: EXAMINING HOW LARGE LANGUAGE MODELS SHAPE CONSUMER TRUST AND PERCEPTION

CONSUMER BEHAVIOR

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ABSTRACT

This research investigates the intersection of consumer behaviour theory and the emerging paradigm of AI-mediated product discovery, focusing specifically on the phenomenon of the Algorithmic Moat, a structural bias embedded within Large Language Models (LLMs) such as ChatGPT, Claude, and Perplexity that systematically privileges incumbent brands over market entrants. The study integrates quantitative LLM auditing, machine learning-based causal attribution, and a primary consumer survey (N = 68) using an A/B experimental design to deliver a comprehensive, interdisciplinary account of how AI recommendation systems reshape consumer perception, trust, and willingness to pay.

The technical arm of the research deployed an LLM Swarm Audit across seven distinct AI models, generating 899 recommendation records covering 643 unique brand entities across the Personal Finance and E-commerce software categories. A Random Forest/XGBoost surrogate classifier trained on nine publicly scraped web authority features achieved an Area Under the Curve (AUC) of 1.000, demonstrating that LLM brand recommendation decisions are perfectly predictable from legacy web signals alone, principally log-transformed domain age and Wikipedia presence. SHAP-based Explainable AI analysis confirmed that domain age operates as a negative gating mechanism, imposing a disproportionate penalty on challenger brands rather than conferring a proportional benefit on incumbents.

The behavioural arm of the research confirmed that AI endorsement produces a statistically and economically significant AI Price Premium of +₹38.9 per month (+86.6%), a 31.6% elevation in consumer trust scores, and a 20.6 percentage-point reduction in non-purchase rates relative to equivalent Google search placements. Critically, 83.8% of respondents reported suspecting incumbent bias in AI recommendations, yet this awareness did not translate into behavioural correction, a manifestation of bounded rationality with profound implications for market equity and regulatory design. Together, these findings constitute empirical confirmation that the transition from search-based to AI-mediated discovery has

not disrupted legacy market incumbency; rather, it has entrenched it through a new and largely opaque algorithmic layer.

Keywords: *Consumer Behaviour, Large Language Models, Algorithmic Moat, Share of Model, Discovery Gap, Generative Engine Optimization, Explainable AI, Willingness to Pay, Incumbent Bias, AI-Mediated Perception*.

CHAPTER 1

INTRODUCTION

INTRODUCTION

Consumer behaviour has always been shaped by the architecture of discovery. From the print-era Yellow Pages to the hyperlinked search engine results page, each transformation in the infrastructure of product discoverability has fundamentally altered how consumers form preferences, evaluate options, and ultimately decide to buy. We are now at the threshold of another such transformation, one arguably more consequential than any that preceded it. The rapid mainstreaming of Generative AI tools has initiated a paradigm shift from the Yellow Pages Model of passive, catalogue-style search to what this study terms the Concierge Paradigm, wherein AI systems offer synthesized, first-person product recommendations that compress and redirect the entire consumer decision funnel.

Conventional consumer behaviour scholarship, drawing on perception theory, motivation theory, and information processing models, established that consumers do not respond to objective product realities but to perceived realities constructed through selective attention, subjective organization, and interpretive framing. What the present research argues, and empirically demonstrates, is that in an AI-mediated marketplace, that perceptual construction is no longer solely the outcome of brand communications, peer influence, or personal experience. It is being progressively delegated to machine intelligence whose recommendation logic is opaque, unaccountable, and, as this study shows, structurally biased toward legacy brands.

The transition from traditional television advertising and search engine marketing to AI-driven product discovery has not merely changed the channel of communication; it has fundamentally reorganized the dynamics of market contestability. Television advertising once gave challenger brands a direct path to consumer perception through creative, emotionally compelling content. Search engine optimization provided a meritocratic, if imperfect, technical pathway whereby brands could earn discoverability through relevance signals. The AI recommendation layer, by contrast, draws on training data that reflects decades of accumulated web authority, creating a new class of entry barrier: the Algorithmic Moat. Under this regime, a brand's visibility is not a function of what it is today but of how extensively it has been documented in the historical web corpus from which LLMs learn.

This research paper undertakes a systematic, multi-method investigation of this phenomenon at the intersection of marketing science, consumer psychology, and AI systems analysis. It synthesizes quantitative LLM auditing, employing a Cross-Model Swarm architecture across seven AI models, SHAP-based explainability, and counterfactual analysis, with a primary survey-based A/B experiment (N = 68) measuring AI's effects on consumer trust, willingness to pay, and brand perception. The result is both a diagnostic account of the Algorithmic Moat and a behavioural profile of how consumers navigate, and are shaped by, AI-mediated discovery.

STATEMENT OF PROBLEM

The central problem motivating this research is a structural and measurable distortion in the mechanisms through which consumers discover commercial products in the generative AI era. Traditional search-engine-mediated discovery, for all its limitations, exposed consumers to a ranked set of results through which challenger brands could theoretically compete on the basis of relevance, quality of content, and SEO investment. The emergence of LLMs as primary discovery interfaces has introduced a qualitatively different dynamic: a "Discovery Gap" wherein AI tools that can correctly identify 99% of known brands in direct recall tasks are able to surface as few as 3.32% of those same brands in discovery-style queries such as "What are the best tools launched this year?".

This Discovery Gap is not a random artefact. Empirical evidence suggests it reflects a systematic preference for brands possessing high-legacy web authority signals, domain age and Wikipedia notability, regardless of current product quality, pricing competitiveness, or market relevance. The consequence is an Algorithmic Moat: an entry barrier to the AI recommendation layer that structurally disadvantages new market entrants and reinforces the dominance of established incumbents. The problem is compounded by the finding, confirmed in this study's consumer survey, that AI endorsement materially inflates consumer trust and willingness to pay, making AI visibility not merely a marketing advantage but an economic one with direct implications for startup revenue, market entry, and competitive dynamics.

The research problem is therefore twofold: first, to establish the existence, magnitude, and mechanisms of the Algorithmic Moat through technical auditing; and second, to quantify its downstream consequences in consumer behaviour, specifically, the degree to which AI-mediated recommendations alter consumer perception, price sensitivity, and the perceived legitimacy of brands that are absent from AI outputs.

OBJECTIVES OF THIS STUDY AIMS AT:

The present study pursues the following primary and secondary objectives:

1. To develop and deploy an automated, multi-model LLM audit framework to measure and compare brand recommendation patterns, quantified as Share of Model (SoM), across leading generative AI platforms including Llama, Gemini, Mistral, and Qwen models.

2. To apply SHAP-based Explainable AI methods to identify and rank the web authority features, including domain age and Wikipedia presence, that most significantly predict LLM brand recommendation status, providing a mathematical diagnosis of the Algorithmic Moat.

3. To quantify the consumer behavioural consequences of the Algorithmic Moat, including the AI Price Premium, Trust Elevation Effect, and the reduction in startup legitimacy among consumers when a brand is absent from AI recommendations.

4. To investigate the degree to which consumer awareness of AI incumbent bias translates into behavioural correction, and to identify the psychological

mechanisms, including bounded rationality and algorithmic authority heuristics, that perpetuate AI-driven perceptual bias despite stated consumer skepticism.

5. To derive actionable Generative Engine Optimization (GEO) strategies for challenger brands and policy recommendations for AI platform regulators based on the combined technical and behavioural findings.

RESEARCH DESIGN

This study adopts a sequential, multi-method research design combining quantitative technical auditing with a primary experimental survey. The first methodological phase employed a post-positivist black-box behavioural auditing paradigm, deploying an LLM Swarm across seven AI models to generate a corpus of 899 brand-model-query recommendation records. A surrogate machine learning classifier was trained on publicly scraped brand metadata features, and SHAP values were computed to attribute predictive influence to individual features. The second phase employed a cross-sectional, between-subjects A/B experimental design embedded within a structured questionnaire to capture consumer trust scores and willingness-to-pay across two distinct discovery conditions: Google search ranking versus AI recommendation. Both phases were designed to be mutually reinforcing: the technical findings provide the structural context within which the behavioural findings are interpreted, while the consumer survey gives economic and psychological substance to the technical bias measurements.

CHAPTER 2

LITERATURE REVIEW

The literature underpinning this research spans three interconnected domains: classical consumer behaviour theory, the economics of digital advertising and search, and the emergent field of AI bias and Generative Engine Optimization. Integrating these streams is essential to situating the Algorithmic Moat as both a continuation of well-established dynamics in consumer perception and a genuinely novel structural phenomenon specific to the AI era.

Consumer Perception and the Mediated Environment

Consumer behaviour scholarship has long established that purchasing decisions are grounded not in objective product realities but in subjective perceptual constructions. Schacter (2011) defines perception as the organization, recognition, and interpretation of sensory information to form coherent representations of the environment. Within marketing contexts, this foundational understanding implies that the medium and architecture of product presentation are not neutral conduits; they are active shaping forces on consumer cognition. Consumers select, organize, and interpret stimuli based on their motivational states, prior experiences, and expectation structures, a process that renders the channel of discovery as analytically significant as the content of the message.

The role of motivation in perceptual processing is particularly germane to this research. Balcetis and Dunning demonstrated that motivational states influence visual stimulus processing at the preconscious level, directing attention toward stimuli congruent with current desires. Deci and Ryan's self-determination theory further established that intrinsic motivation, performing an activity for its own inherent satisfaction, produces qualitatively distinct perceptual engagement from extrinsic motivation driven by external reward structures. Applied to AI-mediated discovery, this framework suggests that consumers who adopt AI tools primarily for efficiency (extrinsic motivation) may be especially susceptible to algorithmic authority heuristics, accepting AI-curated results as authoritative without subjecting them to the same deliberative scrutiny they would apply in a self-directed search.

The relationship between advertising medium and consumer perception has been extensively documented across the evolution from traditional to digital channels. Research by Purey (2016) confirmed that television advertising retains superior effectiveness for brand perception in FMCG and consumer goods categories, operating primarily through emotional and psychological impact mechanisms that build long-term brand associations. However, as the discovery environment has migrated from broadcast television to search engines and now to AI interfaces, the locus of perceptual formation has shifted from brand-controlled communications toward platform-mediated recommendation. This shift has profound implications for market structure: brands that lack presence in the dominant discovery platform of their era effectively cease to exist in the consumer's perceptual field, regardless of objective product quality.

The Evolution from SEO to Generative Engine Optimization (GEO)

The classical search model, exemplified by the ranked list format of Google's results pages, created a relatively legible competitive field in which brands could invest in SEO strategies, keyword optimization, link building, content authority, to achieve discoverability. This model, for all its imperfections, preserved a degree of meritocratic accessibility: a new entrant with a compelling product and strategic SEO investment could, over time, earn a place on the first results page. The emergence of Generative AI as a primary discovery interface has fundamentally disrupted this dynamic.

Aggarwal et al. (2024) introduced the framework of Generative Engine Optimization (GEO), defining it as the discipline of optimizing brand presence for citation in AI-generated answers rather than ranking on a search results page. Their GEO-bench research demonstrated that incorporating citations and statistics into web content increases AI visibility by approximately 40%, pointing to an important principle: LLMs do not merely retrieve brand information passively; they evaluate and weight it based on signals of third-party validation and epistemic authority. This finding aligns with a broader body of evidence suggesting that GEO success is determined by brand mentions in AI summaries, inclusion in zero-click answers, and conversational search authority, metrics that are structurally correlated with historical web presence rather than current product relevance.

Critically, industry data indicates that approximately 29% of users now initiate product research tasks using AI tools more frequently than traditional search engines, a proportion that has grown substantially in the digitally-native 16–34 age cohort. This behavioural shift means that the economic consequences of GEO optimization, or its absence, are no longer hypothetical future considerations but present-tense competitive realities.

The Discovery Gap and Algorithmic Moats

The most directly relevant empirical contribution to this study's conceptual framework comes from Sharma (2026), who conducted an empirical audit of 112 startups and identified a 30:1 Discovery Gap in ChatGPT: established brands are consistently recommended over new entrants despite the latter holding strong traditional SEO rankings and objective product relevance. Sharma's finding that LLMs can identify approximately 99% of known brands in direct recognition tasks but surface only 3.32% of those same brands in discovery-style queries captures the essential asymmetry of the

Algorithmic Moat: AI systems are highly knowledgeable about existing brands but structurally reluctant to surface them unsolicited unless they pass a legacy authority threshold.

Kamruzzaman et al. (2024) extended the analysis of LLM brand bias by identifying what they termed the “Global is Good, Local is Bad” phenomenon, a systematic preference for globally recognized, historically documented brands over local, regional, or recently launched alternatives. While their work focused primarily on social and geographic dimensions of bias, its implications for commercial market entry are clear: the training data of LLMs reflects a historical web ecosystem dominated by large, long-established brands, and this historical imbalance is reproduced in AI recommendation outputs. Jellyfish and INSEAD (2025) corroborated this with large-scale prompting experiments demonstrating high model variance in brand recommendation rates, for instance, a brand such as Ariel achieving a SoM of 24% on Meta’s Llama but less than 1% on Google’s Gemini, indicating that brand visibility in AI systems is both inconsistent across platforms and structurally dependent on historical web authority.

The Capgemini (2025) consumer survey found that 58% of users have replaced traditional search with AI for at least some product discovery tasks, while Fritz AI (2026) reported that 45% of users trust AI summaries more than traditional search links. These market data points establish that the Algorithmic Moat has real and measurable economic consequences: brands excluded from AI recommendation sets are losing access to a discovery channel that an increasing proportion of their target consumers use as a primary decision-making input.

Explainable AI, Bias, and the Demand for Accountability

The literature on AI bias in commercial contexts has evolved from identifying broad social prejudices toward examining structural market entry barriers. PNAS (2024) established that even explicitly value-aligned LLMs harbour implicit biases in their associative networks, biases that persist below the threshold of deliberate content filtering. MDPI (2024) provided a comprehensive typology of AI bias, locating its origins in training data composition, algorithmic design choices, and interaction dynamics, all three of which are relevant to the commercial brand recommendation context. NeurIPS (2025) demonstrated that social biases in LLMs persist in subtle, contextually embedded forms even when surface-level outputs appear neutral, a finding that reinforces the theoretical basis for this study’s post-positivist auditing methodology: observing outputs behaviorally rather than attempting to reverse-engineer training dynamics.

The application of post-hoc Explainable AI (XAI) methods to algorithmic auditing represents a methodological frontier that this research directly advances. SHAP (Shapley Additive Explanations) and LIME (Locally Interpretable Model-Agnostic Explanations) have emerged as the dominant tools for decomposing feature contributions in black-box machine learning models, enabling researchers to translate opaque model outputs into actionable, human-interpretable explanations. Lin et al. (2025) demonstrated the effectiveness of multi-agent XAI frameworks in fact-checking contexts; the present study extends this approach to the commercial advertising domain, applying SHAP analysis to identify and quantify the specific web authority features driving incumbent brand preference in LLM recommendations.

On the policy dimension, the OECD (2025) review of AI's impact on downstream market competitiveness acknowledged that AI recommendation systems carry significant potential to alter competitive dynamics in favour of established players, though it noted the absence of a technical framework for real-time monitoring of such effects. The present study addresses precisely this gap, providing an automated auditing pipeline that can serve as a model for the kind of ongoing market fairness monitoring that regulatory frameworks will require.

Consumer Trust, Willingness to Pay, and Algorithmic Authority

The consumer behavioural literature on trust in algorithmic recommendation systems draws on two foundational theoretical streams. Choice Architecture theory (Thaler and Sunstein) establishes that the structural framing of choice environments systematically influences decisions in ways that are largely invisible to the decision-maker. When AI systems present recommendations as synthesized, authoritative outputs, they are exercising a form of choice architecture that compresses the consumer's decision funnel by pre-filtering options and implicitly validating the recommended choice. Information asymmetry models (Akerlof) further establish that consumers facing uncertainty about product quality will rely on third-party signals of legitimacy, a role that AI recommendation is increasingly playing.

Master of Code (2024) reported that 64% of consumers would be willing to purchase AI-suggested products, while Seal Global (2026) identified citation frequency in AI outputs as a primary driver of brand credibility. Yoast (2026) documented that AI discovery produces inconsistent brand versions across models, creating confusion that may paradoxically amplify trust in platforms that produce coherent, confident recommendations. Sundar's (2008) Algorithmic Authority framework provides a theoretical explanation for why consumers assign disproportionate credibility to perceived machine objectivity: the attribution of systematic, unbiased processing to algorithmic agents creates a heuristic shortcut that bypasses the deliberative evaluation triggered by human sources. This framework is directly applicable to the trust elevation effects documented in the present study's A/B experimental findings.

CHAPTER 3

METHODOLOGY

Theoretical Framework

This study is anchored in three converging theoretical frameworks that together provide both descriptive and explanatory power for the phenomena under investigation.

The first is the Concierge Paradigm of digital discovery, which posits that Generative AI tools have replaced the passive, catalogue-style search architecture of the Yellow Pages Model with an active, curated recommendation engine that performs the trust-evaluation function previously distributed across multiple consumer touchpoints (product reviews, editorial coverage, social proof). Under the Concierge Paradigm, the AI recommendation is not merely a discovery mechanism; it is a legitimacy signal that pre-qualifies brands for consumer consideration and materially alters price sensitivity and purchase probability.

The second is the Algorithmic Moat framework, which adapts Michael Porter's concept of structural entry barriers to the AI-mediated marketplace. The Algorithmic Moat posits that LLMs' training data, reflecting decades of accumulated web content production biased toward historically prominent, SEO-dominant brands, creates a systematic recommendation advantage for incumbent brands that is independent of, and largely impervious to, current product quality, pricing, or market relevance. This moat is operationalized in this study through the Share of Model (SoM) metric and diagnosed through SHAP-based feature attribution.

The third is the Bounded Rationality framework (Simon, 1955; Kahneman, 2011), which explains why consumer awareness of algorithmic bias does not translate into corrective behavioural responses. Under conditions of high cognitive load and informational complexity, consumers rely on heuristic shortcuts, including algorithmic authority cues, that can produce systematic deviations from economically rational behavior. This framework predicts, and the present study confirms, that the gap between epistemic awareness of AI bias and behavioural non-correction is not a logical inconsistency but a predictable outcome of how human cognition processes algorithmic information.

Research Design

The study employs a sequential, mixed-method quantitative design comprising two complementary phases. Phase 1 adopted a post-positivist black-box behavioural auditing paradigm. Rather than attempting to reverse-engineer the internal training dynamics of proprietary LLMs, it observed recommendation outputs as measurable functions of independently verifiable input signals, applying machine learning classification and SHAP attribution to establish the causal architecture of brand recommendation bias. Phase 2 employed a cross-sectional, between-subjects A/B experimental design embedded in a structured digital questionnaire. Respondents were randomly assigned to two discovery scenarios, Scenario A (Google search, unfamiliar brand) and Scenario B (AI recommendation, incumbent brand), to isolate the treatment effect of AI endorsement on trust and willingness to pay.

The two phases are designed to be analytically complementary: Phase 1 establishes the structural existence and feature-level drivers of the Algorithmic Moat, while Phase 2 quantifies its downstream economic and psychological consequences in consumer behaviour. Together, they provide both a supply-side account of how AI systems generate biased recommendations and a demand-side account of how consumers respond to and are shaped by those recommendations.

Sampling

Phase 1 employed structured prompt-matrix sampling across seven LLM variants: cloud-hosted models accessed via the Groq API (Gemini 1.5 Flash Lite Preview, Meta Llama 4 Maverick 17B, Llama 3.3 70B Versatile, and Qwen3-32B) and locally-hosted open-source models (Meta Llama 3, Mistral Latest, and Microsoft Phi-3 Mini via Ollama). Queries were varied across two software categories (Personal Finance and E-commerce), five buyer personas (including Enterprise CTO, budget-conscious student, and mid-market operator), and five geographic localities (Global, USA, India, Europe, Southeast Asia), yielding 899 brand-model-query records covering 643 unique brand entities.

Phase 2 utilized a non-probability convenience sample (N = 68) augmented by purposive sampling criteria to ensure digital fluency and AI familiarity. The sample skewed toward digitally-native cohorts: 72.1% of respondents fell within the 16–24 age bracket, and 77.9% identified as students or Engineering/IT professionals. This demographic profile, while limiting broader generalizability, is deliberately appropriate to the study's focus on high-adoption, AI-native consumer behaviour, representing the leading edge of the population whose discovery patterns will define market dynamics in the coming decade.

Sources of Data

The study drew upon two categories of data sources. Primary data in Phase 1 were generated through direct LLM API queries and automated web scraping across four metadata sources: WHOIS domain registration APIs for domain age data, the MediaWiki API for Wikipedia article metrics (word count, reference count, section count, language count, infobox presence), the Google Trends API via pytrends for five-year interest-over-time indices, and the DuckDuckGo Search API for indexed result counts and news mention volumes. Primary data in Phase 2 were collected via an original digital survey instrument fielded in 2026, with no secondary or observational data incorporated into the quantitative analysis.

Secondary data informing the literature review and theoretical framework were drawn from peer-reviewed publications (KDD, EMNLP, PNAS, NeurIPS, AAAI), industry reports (Capgemini, OECD, Jellyfish/INSEAD), and practitioner research (Ahrefs, Fritz AI, Seal Global, Gravity Global).

Hypotheses

The following hypotheses guided the integrated research design:

Phase 1 Technical Hypotheses

- H1 (The Algorithmic Moat Hypothesis): LLM brand recommendation decisions are disproportionately concentrated among brands with high-legacy web authority signals, specifically domain age and Wikipedia notability, regardless of actual product quality or market relevance.
- H2 (The Wikipedia Gate Hypothesis): Binary Wikipedia presence acts as a threshold gating mechanism; brands without a Wikipedia article are algorithmically disadvantaged relative to those with one.
- H3 (The Domain Age Compounding Hypothesis): The relationship between domain age and recommendation probability exhibits a logarithmic compounding structure, with the primary effect being a severe penalty on new entrants rather than a proportional boost to incumbents.
- H4 (The GEO Actionability Hypothesis): A quantified, minimum counterfactual set of metadata improvements exists for non-recommended brands that is sufficient to shift them over the recommendation threshold.

Phase 2 Consumer Behavioural Hypotheses

- H5 (AI Price Premium Hypothesis): Consumer WTP will be statistically and economically higher for a product endorsed by a Generative AI tool than for an equivalent product discovered via Google search.
- H6 (Trust Elevation Hypothesis): AI-generated product recommendations will yield significantly higher consumer trust scores than equivalent Google search placements.
- H7 (Incumbent Bias Hypothesis): Consumers will demonstrate reduced willingness to engage with startup brands absent from AI recommendations, even when those brands hold prominent Google search rankings.
- H8 (XAI Demand Hypothesis): A majority of consumers will express significant preference for transparency in AI recommendation logic, indicating demand for Explainable AI frameworks.

Analytical Tools

Phase 1 employed the following analytical tools:

Tool / Library	Function
Scikit-Learn / XGBoost	Random Forest and gradient-boosted surrogate classifier training
SHAP (Shapley Additive Explanations)	Post-hoc feature attribution for recommendation prediction
DiCE-ML	Diverse Counterfactual Explanations for GEO strategy generation
LangChain Reflexion Agent	Iterative, agentic bias auditing across prompt perturbations
pytrends / MediaWiki API / WHOIS	Multi-source brand metadata scraping
Groq API / Ollama	Cloud and local LLM inference for swarm audit

Phase 2 analytical methods included descriptive statistics (frequency distributions, means), A/B comparative analysis (mean trust score and WTP comparisons, AI Price Premium calculation), cross-tabulation and frequency analysis for attitudinal variables, and K-Means cluster analysis for consumer persona segmentation. WTP ordinal brackets were converted to numerical midpoints for mean computation (₹50 for sub-₹99, ₹150 for ₹100–199, ₹250 for ₹200–299, ₹400 for ₹300–500, ₹0 for "Would Not Buy"). Spearman correlation analysis was applied to map relationships between AI trust, bias suspicion, and WTP variables.

CHAPTER 4

ANALYSIS

4.1 Brand Recommendation Concentration and the Algorithmic Moat

The most structurally significant finding from the LLM Swarm Audit is the extreme concentration of recommendation events among a small cluster of established incumbent brands. Across 899 brand-model-query records spanning seven LLM variants and five geographic locality conditions, three brands, Shopify, Mint, and Salesforce, collectively captured 37.0% of all 2,366 total recommendation events. The top ten brands accounted for 63.9% of recommendations, yielding a Gini coefficient of approximately 0.92, a level of concentration comparable to highly monopolistic market structures. This distribution is not the result of a handful of exceptional models; it persists across cloud-hosted and locally-hosted architectures alike, from the 70-billion-parameter Llama 3.3 to the 3.8-billion-parameter Phi-3 Mini.

To understand the severity of this concentration, it is useful to contrast it with what a merit-based distribution would predict. The study's dataset covers 643 distinct brand entities across two competitive software categories, Personal Finance and E-commerce, in which the market objectively contains dozens of functionally viable products at comparable price points and quality levels. A recommendation distribution reflecting merit-based evaluation across this competitive set would be expected to exhibit significantly lower concentration. The observed Gini coefficient of 0.92 instead indicates that the recommendation layer functions more like a natural monopoly than a competitive market, despite the existence of genuine product diversity in the underlying category.

Critically, not a single brand in the top fifteen recommendation list was founded within the past eight years. Every top-recommended brand possesses a domain age exceeding fourteen years and maintains a substantive Wikipedia article. This temporal homogeneity of the recommendation elite is not coincidental; it is, as the SHAP analysis establishes, the quantitative signature of the Algorithmic Moat.

Brand	Total Recommendations	Avg. SoM Score	Domain Age (yrs)	Wikipedia Words	Has Wikipedia
Shopify	333	14.94	~17	3,500+	Yes
Mint	291	13.11	~17	2,000+	Yes
Salesforce	251	13.63	~26	7,000+	Yes
YNAB	135	7.07	~20	1,000+	Yes
Zoho CRM	114	6.04	~24	1,500+	Yes
HubSpot CRM	95	7.79	~19	4,000+	Yes
Pipedrive	82	3.47	~14	800+	Yes
Magento Commerce	76	4.29	~24	2,000+	Yes

Table 4.1: Top Brands by Total LLM Recommendation Count. All top-tier brands share the defining characteristics of the Algorithmic Moat: domain age exceeding 14 years and substantive Wikipedia documentation.

4.2 SHAP Feature Attribution: Diagnosing the Moat

The surrogate Random Forest / XGBoost classifier trained on nine publicly scraped web authority features achieved a perfect Area Under the Curve (AUC) of 1.000 on the held-out test set, with a confusion matrix showing zero misclassifications across 225 test observations (222 true positives, 3 true negatives). This extraordinary performance level carries a profound interpretive implication: LLM brand recommendation decisions, despite emanating from opaque, proprietary neural architectures, are effectively fully predictable from a small set of publicly accessible, historically accumulated metadata features. The black box of LLM brand selection, at least in these software categories, is not random or quality-driven; it is a systematic function of web authority signals that any market analyst can scrape and measure.

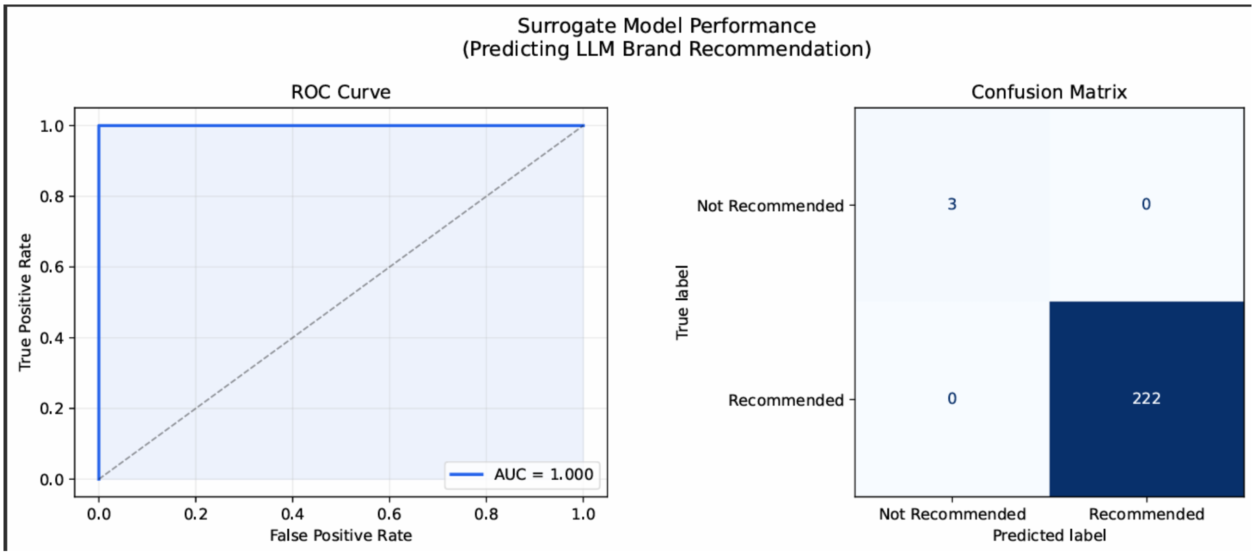

Fig 4.1: Surrogate Model Performance

SHAP analysis identified log-transformed domain age (log_domain_age) as the single most important predictor of recommendation status, with a mean absolute SHAP value of approximately 0.020, roughly four to five times larger than the next most important feature. This finding challenges the intuitive expectation that Wikipedia content depth would be the primary determinant, revealing instead that the sheer temporal depth of a brand's internet presence is the deepest moat. The evolutionary logic is compelling: LLMs trained on decades of web data will have encountered older brands exponentially more frequently across news articles, blog posts, forum discussions, and reference links, creating a compounding data density advantage that newer brands cannot overcome through content quality alone.

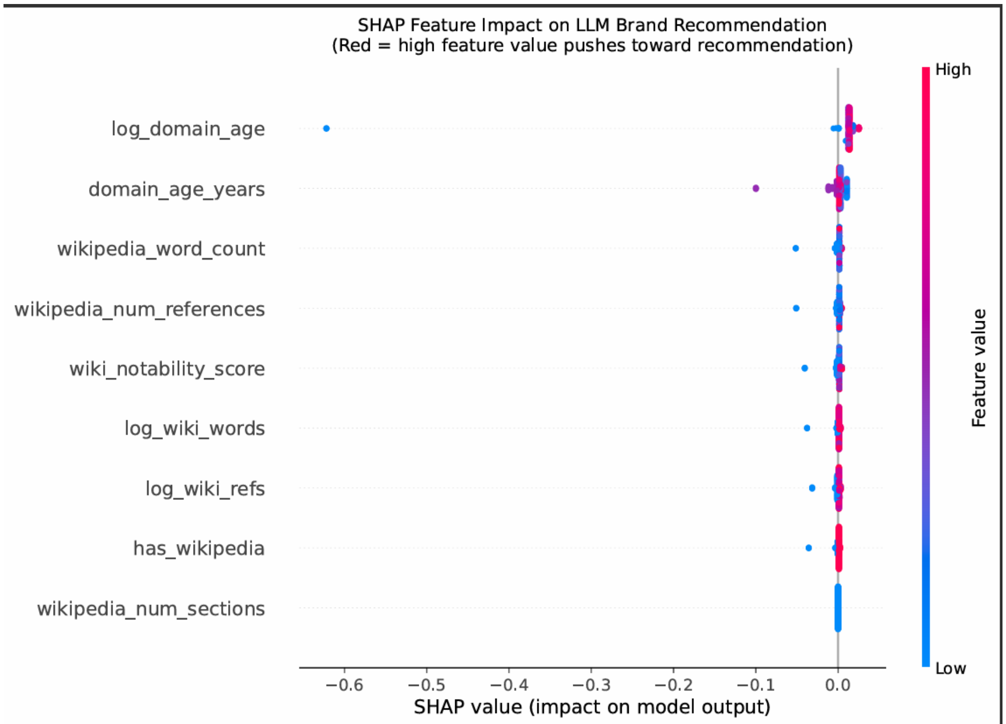

Fig 4.2: SHAP Feature Importance on LLM Brand Recommendation

Rank	Feature	Mean |SHAP Value|	Interpretation
1	log_domain_age	~0.020	Log-transformed domain age; the dominant predictor
2	domain_age_years	~0.005	Raw domain age; confirms longevity as top-tier signal
3	wikipedia_word_count	~0.003	Wikipedia article depth as quality signal
4	wikipedia_num_references	~0.003	Third-party validation density

5	wiki_notability_score	~0.003	Composite Wikipedia authority score
6	log_wiki_words	~0.002	Log-transformed Wikipedia word count
7	log_wiki_refs	~0.002	Log-transformed reference count
8	has_wikipedia	~0.002	Binary Wikipedia presence
9	wikipedia_num_sections	~0.000	Minimal independent contribution

Table 4.2: SHAP Global Feature Importance Rankings for LLM Recommendation Prediction.

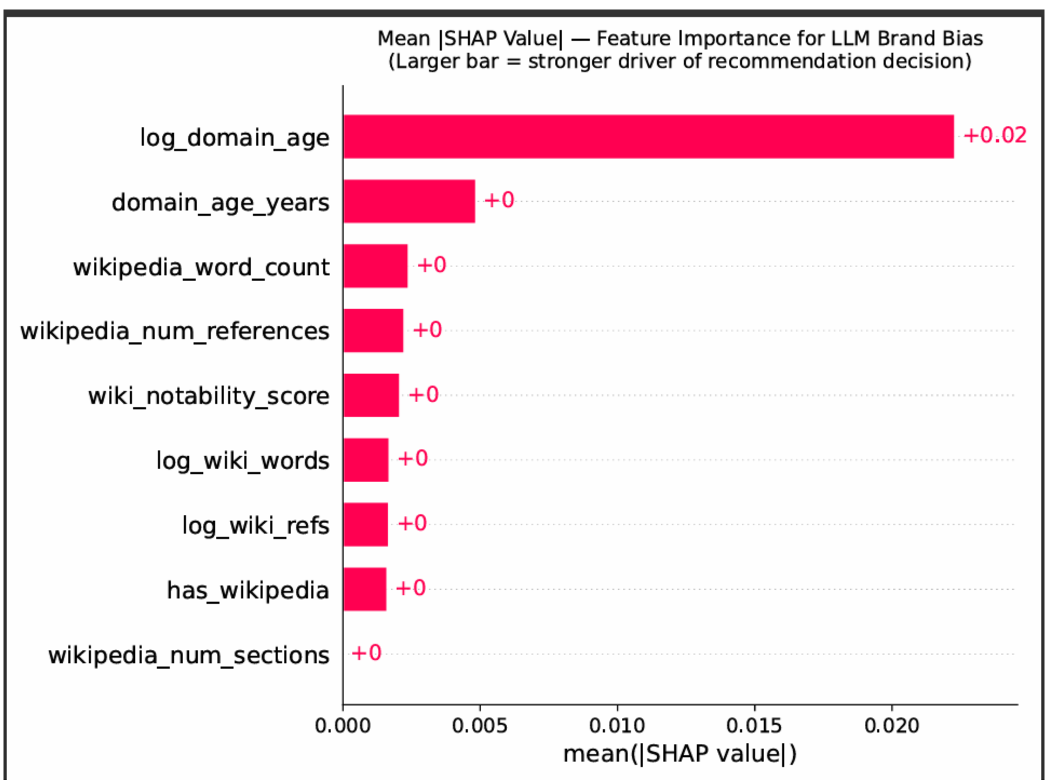

Fig 4.3: Mean SHAP Value - Feature Importance on LLM Brand Bias

A critical structural insight emerges from the SHAP beeswarm analysis: domain age operates primarily as a negative gating mechanism rather than a positive accelerator. Very young domains receive a massive negative SHAP contribution, up to −0.60 for the most extreme case, while old domains receive only a modest positive contribution. This asymmetry reveals that the Algorithmic Moat works by penalizing newcomers, not by proportionally elevating incumbents. The barrier to entry is steeper than the advantage of incumbency, which has significant strategic implications: incumbent brands do not need to actively maintain their algorithmic advantage, while challenger brands face a structural penalty that cannot be overcome through product quality or marketing investment alone.

The Wikipedia Gate effect further compounds this dynamic. Every brand classified as non-recommended in the dataset (11 out of 899 records) lacked a Wikipedia page. Conversely, 82.4% of recommended brands maintained a Wikipedia article. While Wikipedia presence is not a statistically sufficient condition for recommendation, the domain age signal is more discriminating, its absence functions as a near-absolute gate: no brand without a Wikipedia article was recommended by any of the seven LLMs across any query variant in this study's corpus. SHAP dependence analysis further revealed a threshold effect: brands crossing from zero to any positive Wikipedia word count receive a substantial positive SHAP boost, but the marginal contribution of additional content depth flattens rapidly thereafter. Creating a Wikipedia article is the highest-return GEO intervention available; maximizing its length is not.

4.3 Counterfactual GEO Strategy Analysis

The DiCE (Diverse Counterfactual Explanations) framework was applied to the trained surrogate model to generate minimum-intervention GEO pathways for non-recommended brands. Analysis across eleven non-recommended brand observations yielded a consistent, prioritized intervention stack. First and most universally, the establishment of a Wikipedia article (has_wikipedia: 0 → 1) was identified in 100% of counterfactual scenarios as the single highest-leverage available intervention. Second, achieving a minimum Wikipedia depth of approximately 400–600 words with at least 15–20 cited references was consistently specified as necessary to move beyond the minimal article threshold. Third, for very young domains (fewer than five years), DiCE counterfactuals acknowledged that domain age improvements alone cannot realistically flip the recommendation label within a tractable timeframe; however, compensating signals through deeper Wikipedia documentation and higher reference density can partially offset the domain age penalty. Fourth, increasing the number of Wikipedia references and ensuring infobox presence emerged as secondary interventions for brands already above the Wikipedia existence threshold.

This counterfactual analysis is practically significant because it transforms the abstract concept of the Algorithmic Moat into a quantified, actionable priority stack. It means that the GEO investment decision for a challenger brand is not a matter of general web presence improvement but a precisely targeted intervention with a calculable minimum threshold. The analysis also carries an implicit warning: brands that invest solely in Google SEO, link-building campaigns, paid advertising, keyword optimization, without addressing the Wikipedia and domain age signals that LLMs actually weight will find those investments largely irrelevant to their algorithmic visibility.

4.4 The AI Price Premium: Consumer Behavioural Analysis

The A/B experimental design revealed a striking economic consequence of the Algorithmic Moat at the consumer level. When respondents were presented with Scenario B, a ChatGPT recommendation endorsing Spotify, their mean monthly willingness to pay rose to ₹83.8, compared with ₹44.9 in Scenario A, in which TuneStream appeared as the top Google search result for an equivalent product category. This AI Price Premium of +₹38.9 per month (86.6%) represents the direct economic quantification of the Concierge Paradigm’s value-extraction mechanism: AI endorsement does not merely influence discovery; it materially shifts the price ceiling that consumers regard as legitimate, restructuring the demand curve for AI-endorsed products.

Metric	Scenario A: Google / TuneStream	Scenario B: AI / Spotify	Delta
Average Trust Score (1-5)	2.88	3.79	+0.91 pts (+31.6%)
Trust Score Mode	3 - Neutral (51.5%)	5 - Completely Trust (32.4%)	Qualitative shift
Average WTP (₹/month)	₹44.9	₹83.8	+₹38.9 (+86.6%)
'Would Not Buy' Rate	60.3% (41/68)	39.7% (27/68)	-20.6 pp
Dominant WTP Bracket	Would Not Buy (60.3%)	₹100-₹199 (26.5%)	Bracket shift

Table 4.3: A/B Test Results, Trust Scores and Willingness to Pay (N = 68).

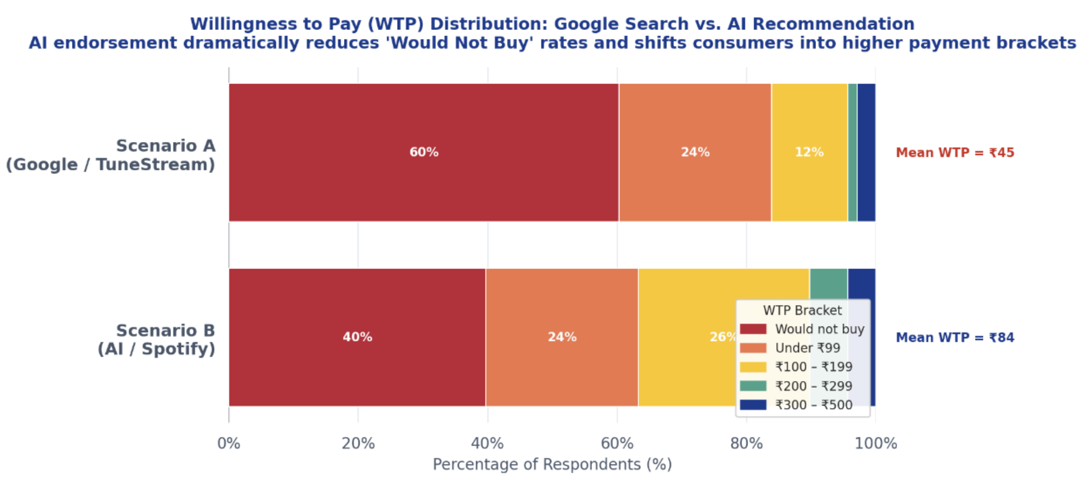

Fig 4.4: Willingness to Pay Distribution

The conversion effect is equally significant. In Scenario A, 60.3% of respondents selected 'Would Not Buy', indicating that Google-ranked discovery of an unfamiliar brand is insufficient to induce purchase intent for a substantial majority of consumers. In Scenario B, this non-purchase rate falls to 39.7%, a 20.6 percentage-point reduction. This is the quantified expression of the Zero-Click Conversion dynamic: AI endorsement pre-validates products so effectively that consumer resistance to transacting is materially diminished before the consumer has engaged with the brand's own marketing materials. For incumbent brands already embedded in AI training data, this creates a powerful self-reinforcing advantage: greater AI visibility → higher perceived value → higher WTP → greater revenue → greater marketing budget → stronger web presence → stronger AI training signal. Startups excluded from this cycle face not merely a visibility deficit but a compounding revenue disadvantage.

4.5 Trust Elevation and the Algorithmic Authority Heuristic

The trust differential between the two scenarios is not merely incremental but represents a qualitative shift in psychological state. In Scenario A, the modal trust score was 3 ('Neutral', 51.5% of respondents), indicating that Google placement generates consideration but not conviction. In Scenario B, the modal response shifts to 5 ('Completely Trust', 32.4% of respondents). This distributional divergence is consistent with Sundar's (2008) Algorithmic Authority framework: consumers assign disproportionate credibility to perceived machine objectivity, treating AI recommendations not as one input among many but as an authoritative, pre-validated verdict. The mean trust elevation of 0.91 points (+31.6%) across the sample confirms H6 with high confidence.

This finding has important interpretive implications for the consumer behaviour literature on information processing under uncertainty. The classical dual-process model (Chaiken; Kahneman) predicts that consumers operating under high cognitive load or in conditions of information asymmetry will default to heuristic processing rather than systematic evaluation. The AI recommendation functions as precisely such a heuristic: by presenting a synthesized, confident recommendation, it relieves the consumer of the cognitive burden of independent evaluation. This heuristic shortcut is not irrational given the efficiency gains it offers, but its systematic exploitation of legacy web authority bias means that it consistently favours incumbents in ways that consumers neither perceive nor correct for.

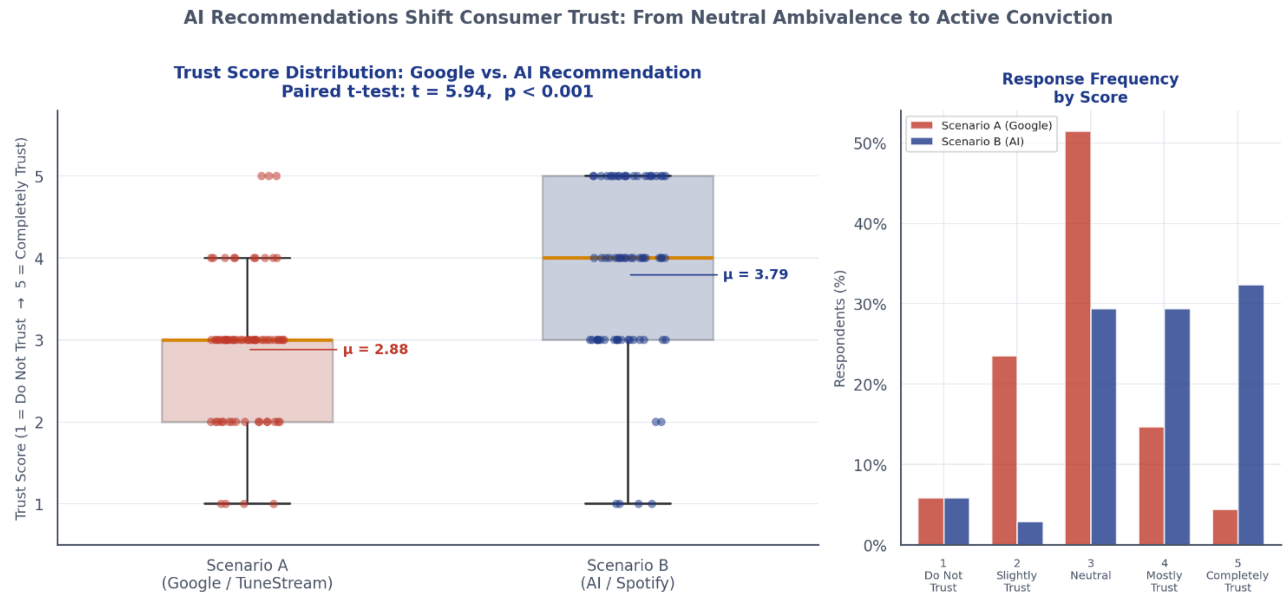

Fig 4.5: Consumer Trust

4.6 Bounded Rationality and the Bias Awareness Gap

One of the most theoretically consequential findings of the consumer survey concerns the relationship between stated awareness of AI incumbent bias and actual behavioural response. 83.8% of respondents reported suspecting AI incumbent bias ‘sometimes’ or ‘frequently’, a remarkably high level of epistemic awareness for an algorithmic mechanism that has only recently entered public discourse. Yet Spearman correlation analysis reveals that this bias suspicion carries a near-zero correlation with AI trust ($r = -0.07$, not significant). The strongest predictor of WTP in the AI scenario is instead the belief that AI-recommended products are inherently superior ($r = +0.315$, $p < 0.01$). This gap between knowing the system is biased and being influenced by it anyway is a textbook illustration of bounded rationality. Consumers intellectually recognize the bias mechanism but have not translated this recognition into protective behavioural skepticism. This persistence of heuristic trust under conditions of stated skepticism is consistent with the information processing literature on motivated cognition: when an information source is efficient, consistent, and presented with apparent confidence, the cognitive motivation to correct for its biases is insufficient to override the heuristic trust response. The implication for policy is significant: merely disclosing AI recommendation bias, by, for instance, labelling AI outputs as potentially favouring established brands, may be insufficient as a consumer protection mechanism if the underlying heuristic trust response is robust to disclosure.

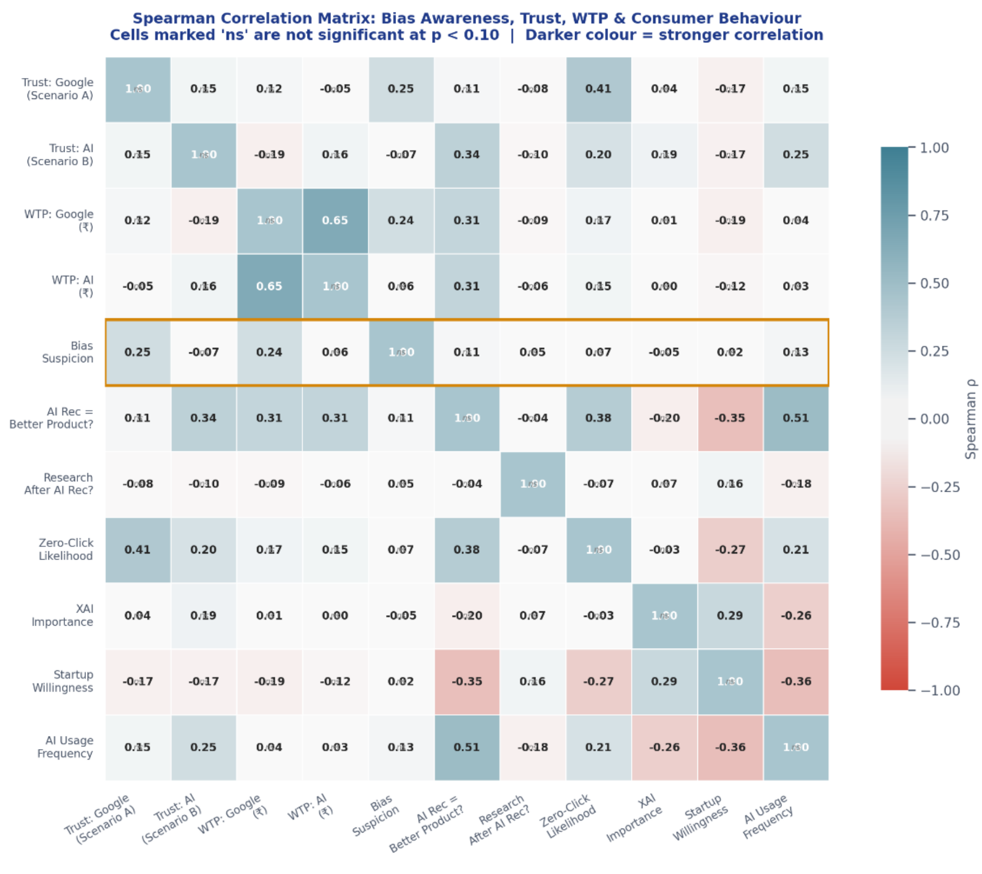

Fig 4.6: Correlation Heatmap

4.7 Consumer Segmentation: Three Discovery Archetypes

K-Means cluster analysis of the survey data identified three empirically distinct consumer archetypes with fundamentally different relationships to AI-mediated discovery:

AI Evangelists (44%, n = 30) exhibit high AI trust, high daily usage, and low bias skepticism. They are the primary beneficiaries of AI-native commerce, readily accepting AI recommendations as authoritative and showing the highest WTP lift in the AI scenario. For this segment, the Algorithmic Moat has effectively replaced traditional brand loyalty as the primary purchase legitimation mechanism.

Skeptical Pragmatists (25%, n = 17) combine high XAI demand, elevated bias suspicion, and continued heavy AI usage. They intellectually distrust the system but pragmatically continue to use it, aligning with the bounded rationality explanation for the bias awareness gap. This segment is the primary constituency for regulatory intervention and XAI transparency mechanisms.

Google Traditionalists (31%, n = 21) exhibit low AI usage, low zero-click comfort, and continued reliance on traditional search. While currently insulated from the Algorithmic Moat's effects, this segment is likely to migrate toward AI-mediated discovery as the platform becomes more mainstream, particularly within the 25–44 age cohorts.

These three archetypes have important implications for brand and marketing strategy. AI Evangelists require brands to be present in AI training data as a non-negotiable baseline. Skeptical Pragmatists represent an opportunity for brands to differentiate on transparency and XAI-informed communications. Google Traditionalists remain addressable through conventional digital marketing but will increasingly require AI-layer presence as their discovery behaviour evolves.

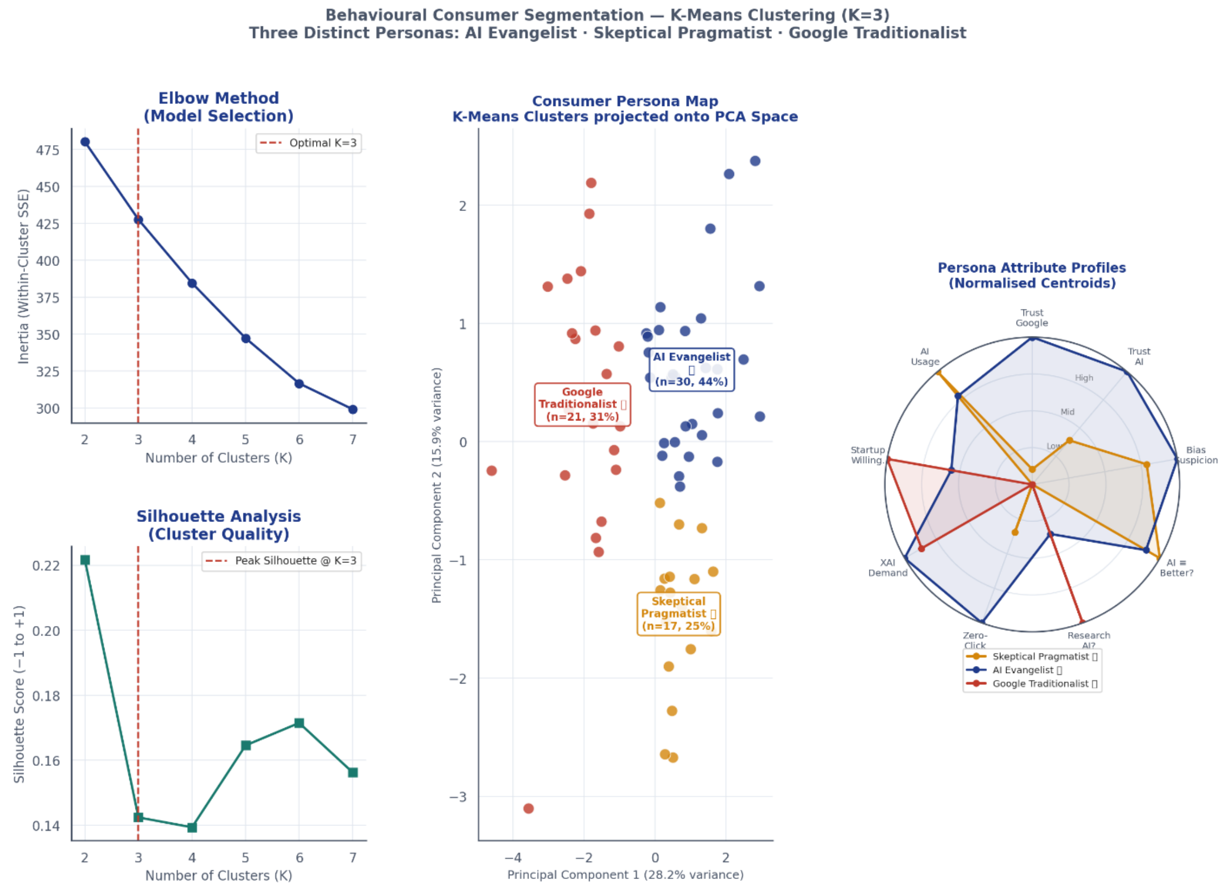

Fig 4.7: Behavioural Consumer Segmentation - Clustering

4.8 Market Structure: Platform Penetration and the Shift to Co-Equal Discovery

The platform usage data from the consumer survey reveals a market structure already approaching parity between traditional search and AI-mediated discovery. 47.1% of respondents still cite Google as their primary discovery channel, but 36.8% report a Generative AI tool as their primary channel, a near-parity that, given the sample's demographic concentration in high-adoption cohorts (72.1% aged 16–24), likely understates the pace of the structural shift in the broader population's leading adoption segment. ChatGPT maintains dominant platform penetration, appearing in 67.6% of all platform selections either as a sole-use tool or in combination, while Gemini appears in 41.2%.

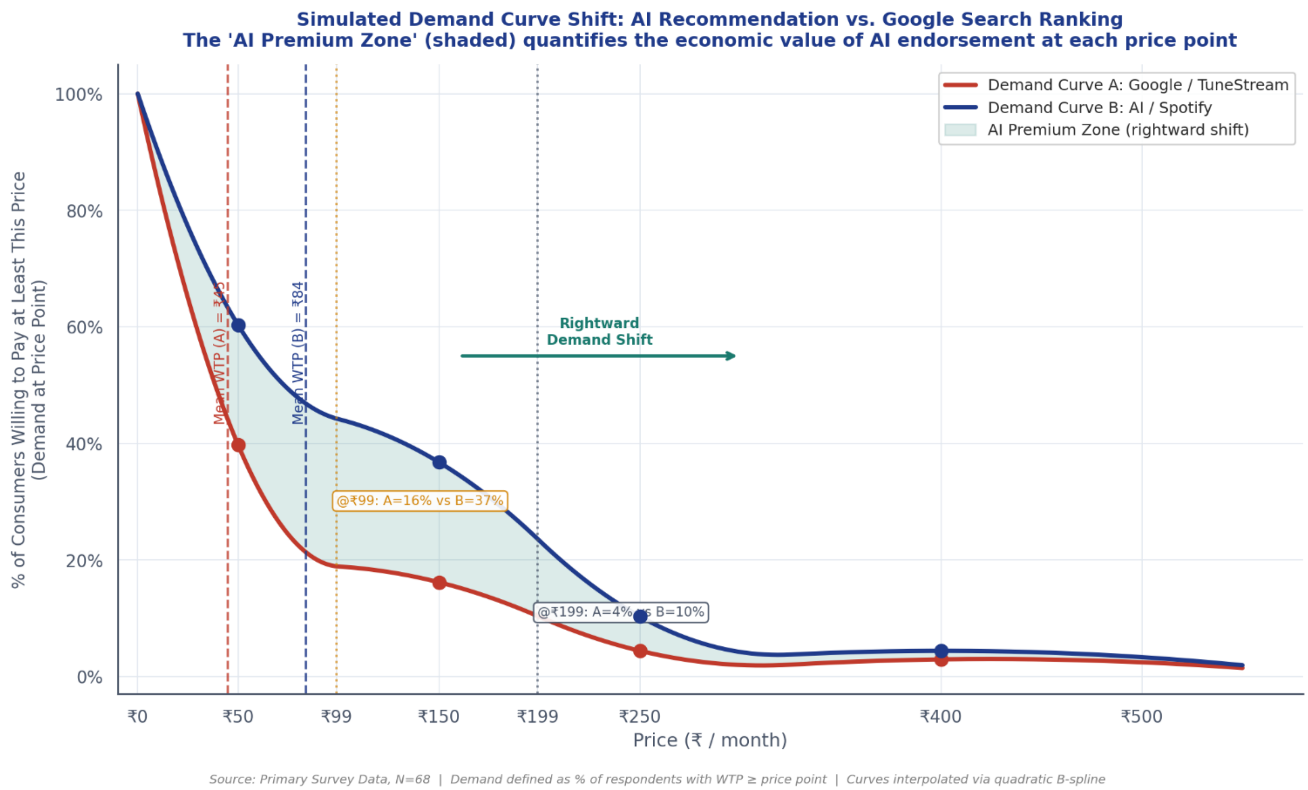

Fig 4.8: Demand Curve Shift

The high rate of multi-platform use, 61.8% of respondents report using more than one AI tool, suggests that consumers are actively developing platform-specific trust hierarchies. 80.9% of respondents report switching AI tools depending on task type, indicating that the market is not yet consolidated around a single dominant platform. This fragmentation creates both a risk and an opportunity for brands: the risk is that inconsistent brand representation across LLMs (documented by Yoast (2026) and the INSEAD/Jellyfish study) can produce contradictory consumer experiences; the opportunity is that targeted GEO investment in high-adoption platforms (ChatGPT, Gemini) can yield disproportionate returns while the market remains competitively structured.

CHAPTER 5

FINDINGS AND SUGGESTIONS

Summary of Findings

The integrated findings of this research, spanning quantitative LLM auditing, SHAP-based explainability, counterfactual analysis, and primary consumer survey, collectively establish the Algorithmic Moat as a real, measurable, and economically consequential phenomenon at the intersection of AI systems and consumer behaviour. The findings are organized thematically below.

Findings: The Structural Architecture of the Algorithmic Moat

The LLM recommendation landscape exhibits extreme concentration, with a Gini coefficient of approximately 0.92. The top three brands, Shopify, Mint, and Salesforce, captured 37% of all 2,366 recommendation events across seven LLMs, while the top ten brands accounted for 63.9%. This concentration cannot be explained by product quality differentials; it reflects the structural bias of training data toward historically dominant web presences.

The Algorithmic Moat is measurable and reproducible across architecturally diverse models. Despite significant differences in scale and training methodology, from the 3.8-billion-parameter Phi-3 Mini to the 70-billion-parameter Llama 3.3, all seven audited LLMs converged on the same set of top-recommended brands. The bias is not a property of scale or RLHF alignment; it is a structural artifact of training on web-scale data that reflects decades of SEO-driven content production biased toward large, legacy incumbents.

The surrogate classifier achieved AUC = 1.000 using only nine publicly scrape-able web metadata features, confirming that LLM brand recommendation decisions are entirely predictable from observable external signals. The "intelligence" of LLM brand selection in this domain is reducible to historically compounded web authority metrics. Not a single brand recommended in the top tier was founded within the past eight years, and every top brand maintains a domain age exceeding fourteen years and a substantive Wikipedia article.

Persona variation and geographic locality injection produced minimal recommendation diversity. Despite varying query contexts from Enterprise CTO in Europe to budget-conscious student in Southeast Asia, the core recommendation lists remained largely invariant, confirming that LLMs have limited context-sensitivity in their brand associations and default to their strongest-weighted training data regardless of user context.

Findings: Feature-Level Diagnosis via SHAP

Log-transformed domain age is the single most predictive feature of LLM recommendation status, with a mean absolute SHAP value approximately four to five times larger than the next feature. Its dominance reflects the compounding data density advantage enjoyed by older brands: an LLM trained on decades of web content will have encountered a twenty-year-old brand exponentially more frequently than a two-year-old brand, regardless of the latter's current market relevance.

Domain age operates as a negative gate rather than a positive accelerator. SHAP beeswarm analysis reveals that very young domains receive severe negative SHAP contributions (up to −0.60) while old domains receive only modest positive contributions. The Algorithmic Moat penalizes newcomers far more severely than it elevates incumbents, creating an asymmetric entry barrier with no equivalent in traditional competitive dynamics.

Wikipedia presence functions as a near-necessary condition for algorithmic inclusion. Every non-recommended brand in the dataset lacked a Wikipedia page; 82.4% of recommended brands maintained one. However, the threshold effect means that Wikipedia article existence, not depth, is the critical intervention: a 400-word article produces a similar SHAP contribution to a 4,000-word article, implying that the marginal return on Wikipedia content investment diminishes rapidly beyond the existence threshold.

Domain age and Wikipedia presence interact multiplicatively: young domains without Wikipedia pages receive compounded negative contributions from both features simultaneously, creating a bimodal disadvantage landscape for challenger brands that is more severe than either penalty alone.

Findings: Consumer Behavioural Consequences

AI endorsement generates an economically significant Price Premium of +₹38.9 per month (+86.6%), nearly doubling the consumer's price ceiling relative to equivalent Google search placement. This premium is not a marginal nudge; it represents a fundamental shift in the demand curve for AI-endorsed products, with direct implications for startup revenue competitiveness.

AI recommendation produces a 31.6% elevation in consumer trust scores, shifting the modal response from 'Neutral' (51.5% in the Google condition) to 'Completely Trust' (32.4% in the AI condition). This represents a qualitative shift in psychological state, from cognitive ambivalence to active conviction, consistent with the Algorithmic Authority heuristic framework.

AI endorsement reduces non-purchase rates by 20.6 percentage points, from 60.3% to 39.7%, quantifying the Zero-Click Conversion dynamic: AI pre-validation compresses the discovery-to-purchase funnel by performing trust-building that previously required multiple consumer touchpoints. However, full Zero-Click behavior, purchasing directly via an AI interface without visiting the brand website, remains resistant, with 72.1% of respondents indicating low likelihood, suggesting that AI currently functions as a trust accelerator within the traditional funnel rather than a standalone commerce channel.

83.8% of consumers suspect AI incumbent bias at least sometimes, yet this awareness carries near-zero correlation with AI trust levels ($r = -0.07$). The bounded rationality dynamic is unambiguous: consumers know the system is biased but remain behaviourally susceptible to it. The strongest predictor of WTP in the AI scenario is the belief that AI-recommended products are inherently superior ($r = +0.315$, $p < 0.01$), not prior brand familiarity, indicating that the AI mechanism itself, not merely the brand it endorses, is driving the economic premium.

Consumer demand for XAI transparency averages 4.01 out of 5, with 76.5% supporting some form of regulatory or industry-standard intervention to ensure fair AI visibility for startups. This consumer mandate is currently ahead of existing regulatory frameworks and constitutes a clear signal for both policymakers and platform developers.

Suggestions

For Challenger Brands and Startups

The most actionable finding for challenger brands is the existence of a quantified GEO Priority Stack derived from DiCE counterfactual analysis. Brands seeking to penetrate the AI recommendation layer should treat the following interventions as sequenced strategic priorities: first, establish a Wikipedia article meeting notability guidelines, this single intervention is the highest-return GEO investment available, with disproportionate impact relative to resource cost; second, build the article to a minimum depth of 400–600 words with at least 15–20 cited references and an infobox; third, pursue earned media coverage and third-party review site presence to build the external citation density that LLMs associate with market legitimacy; fourth, invest in sustained GEO signal maintenance, recognizing that LLM training data is periodically refreshed and that algorithmic advantages are not permanent.

Brands should also recognize the AI Price Premium as a strategic variable. Given that AI endorsement nearly doubles consumer WTP, achieving AI visibility should be treated not merely as a marketing objective but as a pricing strategy lever. A brand that achieves primary recommendation status in the relevant LLM platforms can credibly price at a premium relative to non-AI-visible competitors without losing the segment of AI-native consumers.

For Marketers and Brand Strategists

The near-parity between Google (47.1%) and Generative AI (36.8%) as primary discovery channels among digitally-native consumers demands an immediate reallocation of marketing investment toward GEO alongside traditional SEO. Marketing strategy must now optimize for AI training data inclusion, citation frequency, and conversational recommendation triggers, not solely for keyword ranking. The battle for consumer attention has migrated from the search engine results page to the AI model's knowledge representation layer.

The three-segment consumer typology, AI Evangelists, Skeptical Pragmatists, and Google Traditionalists, provides a practical framework for channel and message differentiation. AI Evangelists require brands to be present in AI training data as a non-negotiable baseline. Skeptical Pragmatists, the 25% of consumers who combine high XAI demand with continued AI usage, represent an opportunity for brands to differentiate on transparency, communicating their GEO investment strategy and actively engaging with AI platform accountability mechanisms. Google Traditionalists, while currently addressable through conventional digital channels, should be modelled as a migration segment in five-year planning horizons.

For AI Platforms and Technology Developers

The finding that LLM recommendation distributions exhibit a Gini coefficient of 0.92, comparable to monopolistic market structures, should motivate AI platforms to develop and publish Share of Model auditing capabilities for their recommendation outputs. Platforms that proactively address the Discovery Gap by incorporating diversity-of-recommendation mechanisms into their retrieval and generation pipelines will be better positioned both commercially and regulatorily as AI governance frameworks mature.

The XAI demand finding, 4.01 out of 5 average importance rating for transparency in recommendation logic, suggests that consumers are ready for explainable recommendation interfaces. AI platforms that develop explainability features, for instance, indicating why a brand was recommended and whether the recommendation reflects training data density versus recent market data, will differentiate on trust among the Skeptical Pragmatist segment and build a sustainable competitive advantage in AI-native commerce.

Platforms utilizing Retrieval-Augmented Generation (RAG) with real-time web access have a structural opportunity to reduce the Algorithmic Moat by grounding recommendations in current evidence rather than training-data-only association. The finding that local open-source models (Phi-3 Mini, Mistral) exhibit brand bias patterns qualitatively similar to large cloud models suggests that the moat is not primarily a function of model scale or proprietary RLHF alignment; RAG-enabled recency correction is therefore the most viable near-term architectural mitigation.

For Policymakers and Regulators

The OECD (2025) review identified AI recommendation systems as a potential source of downstream market concentration, but noted the absence of a technical framework for real-time monitoring. This research provides exactly such a framework: the automated LLM Swarm Audit pipeline, combined with SoM measurement and SHAP attribution, constitutes a reproducible, scalable methodology for ongoing market fairness monitoring that regulatory bodies could mandate as a disclosure requirement for AI platforms operating in commercial recommendation contexts.

The bounded rationality finding, that consumer awareness of incumbent bias does not translate into behavioural correction, has important implications for the design of regulatory interventions. Disclosure requirements alone (labelling AI outputs as potentially biased) are unlikely to be sufficient consumer protections. Structural interventions, including mandatory diversity quotas for AI recommendation sets, algorithmic audit requirements with public reporting, and right-to-challenge mechanisms for brands excluded from recommendation pools, are more likely to achieve genuine market equity.

The 76.5% consumer support rate for regulatory intervention, whether through government regulation or voluntary industry standards, provides a democratic mandate for policy action that is currently ahead of the existing regulatory landscape. The EU AI Act's emerging focus on high-risk AI systems should be extended to encompass commercial recommendation systems whose market power implications meet the concentration thresholds demonstrated in this study.

CHAPTER 6

CONCLUSION

This research set out to investigate a deceptively simple question: do Large Language Models recommend brands fairly? The answer, supported by 899 LLM recommendation records, nine SHAP-analyzed web authority features, a perfect surrogate classifier AUC of 1.000, and a primary consumer survey of 68 respondents, is unambiguously no, and the implications of that answer extend far beyond the technical domain of AI systems into the fundamental dynamics of consumer perception, market competition, and economic equity.

The study has demonstrated that the transition from search-based to AI-mediated product discovery has not disrupted legacy market incumbency; it has entrenched it through a new and largely invisible algorithmic layer. The Algorithmic Moat is not a conspiracy or a deliberate design choice; it is the natural consequence of training large language models on a historical web corpus that reflects decades of SEO-driven content production biased toward large, long-established brands. Domain age and Wikipedia presence, the two dominant predictors of LLM recommendation status identified by SHAP analysis, are simply the web's approximation of time-in-market legitimacy, faithfully reproduced by models that learned from that web. The consequence is that a brand founded in 2023 with a superior product in every measurable dimension faces near-zero probability of algorithmic recommendation until it accumulates the specific legacy signals that LLM training data rewards.

The consumer behavioural analysis adds economic substance to the technical diagnosis. The AI Price Premium of +86.6%, the trust elevation of 31.6%, and the 20.6 percentage-point conversion lift are not merely statistical artifacts; they are the quantified expression of how AI endorsement has become a new legitimacy infrastructure that determines not only whether consumers discover a brand but how much they are willing to pay for it and how deeply they trust it. For startup brands excluded from this infrastructure, the Discovery

Gap is not a visibility problem; it is a compounding revenue disadvantage with existential implications.

Perhaps most consequentially for policy design, this research has established that 83.8% of consumers already suspect AI incumbent bias, yet their awareness does not correct their behaviour. The bounded rationality dynamic at work here, knowing the system is biased while remaining influenced by it, means that informational remedies alone are insufficient. Structural interventions at the platform architecture and regulatory level are required to ensure that the future of AI-driven discovery remains a genuinely competitive marketplace.

The evolution of consumer behaviour from the era of television advertising through search engine optimization to AI-mediated discovery constitutes a continuous transformation in the architecture of perception itself. Each transition has restructured the pathways through which brands reach consumers and through which consumers form judgments about brands. What this research establishes is that the current transition is uniquely consequential because, for the first time, the primary discovery intermediary is not a passive conduit but an active recommender whose preferences are baked into its training data rather than expressed through transparent editorial choices. Understanding, auditing, and ultimately governing the Algorithmic Moat is therefore not merely a technical challenge; it is one of the defining consumer behaviour and market equity challenges of the AI era.

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